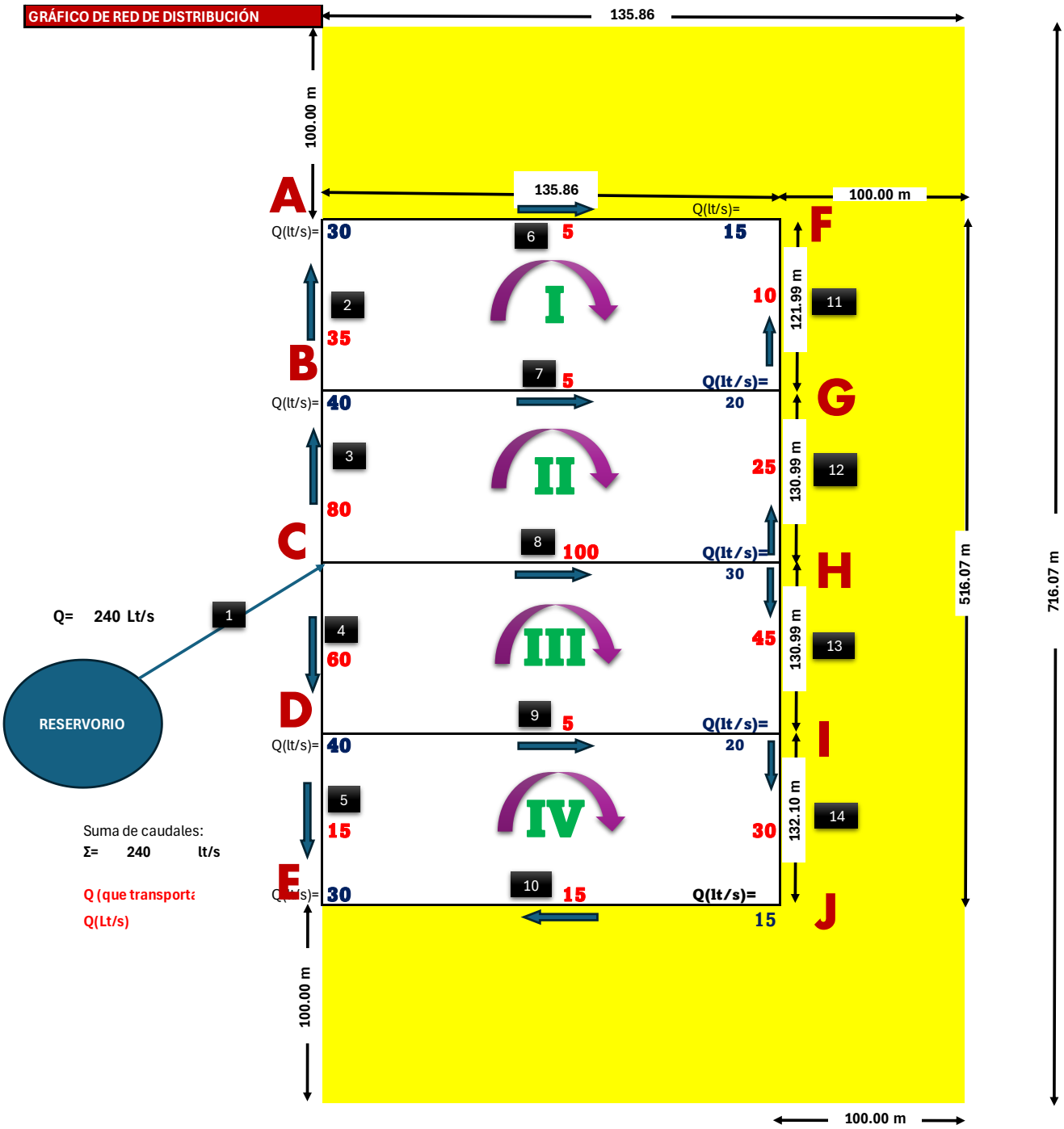
CÁLCULO RED DE DISTRIBUCIÓN CERRADA

PROYECTO: "ABASTECIMIENTO DE AGUA POTABLE A MARGEN IZQUIERDO DE LA UNIVERSIDAD PERUANA UNIÓN, SEDE JULIACA, SAN ROMÁN - PUNO"

GRÁFICO DE RED DE DISTRIBUCIÓN



CÁLCULO HIDRÁULICO DE LA RED

MÉTODO HARDY CROSS

DENSIDAD POBLACIONAL	2770	hab / ha
DOTACIÓN	200	Lt / hab / dia
COEFICIENTE VARIACIÓN	1.8	.=K2.
PVC "C"	150	
n=	1.852	Hazen Williams

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ITERACIÓN I

ANILLO	RUTA		TRAMO	LONG(m)	Diam(pulg.)	Diam(m)	caudal q(lt/s)	coef. K	perdida de carga hf(m)	hf/Q	q(lt/s)	Caudal corregido Q(lt/s)
I	B	A	2	121.99	8	0.20	35	284.96	0.5733	0.0164	-0.5400	34.4600
	A	F	6	135.86	6	0.15	5	1288.24	0.0705	0.0141	-0.5400	4.4600
	B	G	7	135.86	6	0.15	-5	1288.24	-0.0705	0.0141	-0.6955	-5.6955
	F	G	11	121.99	6	0.15	-10	1156.73	-0.2287	0.0229	-0.5400	-10.5400
II	C	B	3	121.99	8	0.20	80	284.96	2.6503	0.0331	0.1555	80.1555
	B	G	7	135.86	6	0.15	5	1288.24	0.0705	0.0141	0.6955	5.6955
	C	H	8	135.86	10	0.25	-100	107.05	-1.5052	0.0151	14.2403	-85.7597
	H	G	12	121.99	6	0.15	-25	1156.73	-1.2480	0.0499	0.1555	-24.8445
III	C	D	4	121.99	8	0.20	-60	284.96	-1.5557	0.0259	-14.0848	-74.0848
	C	H	8	135.86	10	0.25	100	107.05	1.5052	0.0151	-14.2403	85.7597
	D	I	9	135.86	6	0.15	-5	1288.24	-0.0705	0.0141	-3.6893	-8.6893
	H	I	13	121.99	6	0.15	45	1156.73	3.7066	0.0824	-14.0848	30.9152
IV	D	E	5	121.99	8	0.20	-15	284.96	-0.1194	0.0080	-10.3955	-25.3955
	D	I	9	135.86	6	0.15	5	1288.24	0.0705	0.0141	3.6893	8.6893
	E	J	10	135.86	6	0.15	15	1288.24	0.5397	0.0360	-10.3955	4.6045
	J	I	14	121.99	6	0.15	30	1156.73	1.7493	0.0583	-10.3955	19.6045

ANILLO I			Σhf<0.1
Σhf=	0.0675	m	SI CUMPLE
Σhf/q=	0.0675	m/l/s	
q=	-0.5399568	l/s	

ANILLO II			Σhf<0.1
Σhf=	-0.0323	m	SI CUMPLE
Σhf/q=	0.1122	m/l/s	
q=	0.15549799	l/s	

ANILLO III			Σhf<0.1
Σhf=	3.5856	m	NO CUMPLE
Σhf/q=	0.1375	m/l/s	
q=	-14.0847935	l/s	

ANILLO IV			Σhf<0.1
Σhf=	2.2401	m	NO CUMPLE
Σhf/q=	0.1164	m/l/s	
q=	-10.3955278	l/s	

ITERACIÓN II

ANILLO	RUTA		TRAMO	LONG(m)	Diam(pulg.)	Diam(m)	caudal q(lt/s)	coef. K	perdida de carga hf(m)	hf/Q	q(lt/s)	Caudal corregido Q(lt/s)
I	B	A	2	121.99	8	0.20	34.4600	284.96	0.5570	0.0162	-0.5400	33.9201
	A	F	6	135.86	6	0.15	4.4600	1288.24	0.0571	0.0128	-0.5400	3.9201
	B	G	7	135.86	6	0.15	-5.6955	1288.24	-0.0898	0.0158	1.3122	-4.3833
	F	G	11	121.99	6	0.15	-10.5400	1156.73	-0.2521	0.0239	-0.5400	-11.0799
II	C	B	3	121.99	8	0.20	80.1555	284.96	2.6599	0.0332	-1.8521	78.3034
	B	G	7	135.86	6	0.15	5.6955	1288.24	0.0898	0.0158	-1.3122	4.3833
	C	H	8	135.86	10	0.25	-85.7597	107.05	-1.1325	0.0132	0.2227	-85.5370
	H	G	12	121.99	6	0.15	-24.8445	1156.73	-1.2337	0.0497	-1.8521	-26.6966
III	C	D	4	121.99	8	0.20	-74.0848	284.96	-2.2989	0.0310	-2.0748	-76.1596
	C	H	8	135.86	10	0.25	85.7597	107.05	1.1325	0.0132	-0.2227	85.5370
	D	I	9	135.86	6	0.15	-8.6893	1288.24	-0.1963	0.0226	2.4006	-6.2887
	H	I	13	121.99	6	0.15	30.9152	1156.73	1.8494	0.0598	-2.0748	28.8404
IV	D	E	5	121.99	8	0.20	-25.3955	284.96	-0.3165	0.0125	-4.4754	-29.8709
	D	I	9	135.86	6	0.15	8.6893	1288.24	0.1963	0.0226	-2.4006	6.2887
	E	J	10	135.86	6	0.15	4.6045	1288.24	0.0606	0.0132	-4.4754	0.1291
	J	I	14	121.99	6	0.15	19.6045	1156.73	0.7956	0.0406	-4.4754	15.1291

ANILLO I			Σhf<0.1
Σhf=	0.0686	m	SI CUMPLE
Σhf/q=	0.0686	m/l/s	
q=	-0.5399568	l/s	

ANILLO II			Σhf<0.1
Σhf=	0.3835	m	NO CUMPLE
Σhf/q=	0.1118	m/l/s	
q=	-1.85211779	l/s	

ANILLO III			Σhf<0.1
Σhf=	0.4867	m	NO CUMPLE
Σhf/q=	0.1267	m/l/s	
q=	-2.0748107	l/s	

ANILLO IV			Σhf<0.1
Σhf=	0.7359	m	NO CUMPLE
Σhf/q=	0.0888	m/l/s	
q=	-4.47539151	l/s	

ITERACIÓN III

ANILLO	RUTA		TRAMO	LONG(m)	Diam(pulg.)	Diam(m)	caudal q(lt/s)	coef. K	perdida de carga hf(m)	hf/Q	q(lt/s)	Caudal corregido Q(lt/s)
I	B	A	2	121.99	8	0.20	33.9201	284.96	0.5410	0.0159	-0.5400	33.3801
	A	F	6	135.86	6	0.15	3.9201	1288.24	0.0450	0.0115	-0.5400	3.3801
	B	G	7	135.86	6	0.15	-4.3833	1288.24	-0.0553	0.0126	-0.2190	-4.6023
	F	G	11	121.99	6	0.15	-11.0799	1156.73	-0.2765	0.0250	-0.5400	-11.6199
II	C	B	3	121.99	8	0.20	78.3034	284.96	2.5472	0.0325	-0.3209	77.9825
	B	G	7	135.86	6	0.15	4.3833	1288.24	0.0553	0.0126	0.2190	4.6023
	C	H	8	135.86	10	0.25	-85.5370	107.05	-1.1271	0.0132	0.7079	-84.8291
	H	G	12	121.99	6	0.15	-26.6966	1156.73	-1.4094	0.0528	-0.3209	-27.0175
III	C	D	4	121.99	8	0.20	-76.1596	284.96	-2.4195	0.0318	-1.0288	-77.1884
	C	H	8	135.86	10	0.25	85.5370	107.05	1.1271	0.0132	-0.7079	84.8291
	D	I	9	135.86	6	0.15	-6.2887	1288.24	-0.1079	0.0172	0.4145	-5.8741
	H	I	13	121.99	6	0.15	28.8404	1156.73	1.6261	0.0564	-1.0288	27.8116
IV	D	E	5	121.99	8	0.20	-29.8709	284.96	-0.4275	0.0143	-1.4434	-31.3143
	D	I	9	135.86	6	0.15	6.2887	1288.24	0.1079	0.0172	-0.4145	5.8741
	E	J	10	135.86	6	0.15	0.1291	1288.24	0.0001	0.0006	-1.4434	-1.3143
	J	I	14	121.99	6	0.15	15.1291	1156.73	0.4923	0.0325	-1.4434	13.6857

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ANILLO I			Σhf<0.1
Σhf=	0.0650	m	SI CUMPLE
Σhf/q=	0.0650	m/l/s	
q=	-0.5399568	l/s	

ANILLO II			Σhf<0.1
Σhf=	0.0660	m	SI CUMPLE
Σhf/q=	0.1111	m/l/s	
q=	-0.32092911	l/s	

ANILLO III			Σhf<0.1
Σhf=	0.2258	m	NO CUMPLE
Σhf/q=	0.1185	m/l/s	
q=	-1.02881628	l/s	

ANILLO IV			Σhf<0.1
Σhf=	0.1728	m	NO CUMPLE
Σhf/q=	0.0646	m/l/s	
q=	-1.44336483	l/s	

ITERACIÓN IV

ANILLO	RUTA		TRAMO	LONG(m)	Diam(pulg.)	Diam(m)	caudal q(lt/s)	coef. K	perdida de carga hf(m)	hf/Q	q(lt/s)	Caudal corregido Q(lt/s)
I	B	A	2	121.99	8	0.20	33.3801	284.96	0.5251	0.0157	-0.5400	32.8402
	A	F	6	135.86	6	0.15	3.3801	1288.24	0.0342	0.0101	-0.5400	2.8402
	B	G	7	135.86	6	0.15	-4.6023	1288.24	-0.0605	0.0131	-0.3585	-4.9608
	F	G	11	121.99	6	0.15	-11.6199	1156.73	-0.3020	0.0260	-0.5400	-12.1598
II	C	B	3	121.99	8	0.20	77.9825	284.96	2.5279	0.0324	-0.1815	77.8010
	B	G	7	135.86	6	0.15	4.6023	1288.24	0.0605	0.0131	0.3585	4.9608
	C	H	8	135.86	10	0.25	-84.8291	107.05	-1.1098	0.0131	0.0728	-84.7563
	H	G	12	121.99	6	0.15	-27.0175	1156.73	-1.4409	0.0533	-0.1815	-27.1990
III	C	D	4	121.99	8	0.20	-77.1884	284.96	-2.4804	0.0321	-0.2543	-77.4427
	C	H	8	135.86	10	0.25	84.8291	107.05	1.1098	0.0131	-0.0728	84.7563
	D	I	9	135.86	6	0.15	-5.8741	1288.24	-0.0951	0.0162	0.1605	-5.7137
	H	I	13	121.99	6	0.15	27.8116	1156.73	1.5203	0.0547	-0.2543	27.5573
IV	D	E	5	121.99	8	0.20	-31.3143	284.96	-0.4665	0.0149	-0.4148	-31.7291
	D	I	9	135.86	6	0.15	5.8741	1288.24	0.0951	0.0162	-0.1605	5.7137
	E	J	10	135.86	6	0.15	-1.3143	1288.24	0.0059	-0.0045	-0.4148	-1.7291
	J	I	14	121.99	6	0.15	13.6857	1156.73	0.4089	0.0299	-0.4148	13.2709

ANILLO I			Σhf<0.1
Σhf=	0.0650	m	SI CUMPLE
Σhf/q=	0.0650	m/l/s	
q=	-0.5399568	l/s	

ANILLO II			Σhf<0.1
Σhf=	0.0376	m	SI CUMPLE
Σhf/q=	0.1120	m/l/s	
q=	-0.18149298	l/s	

ANILLO III			Σhf<0.1
Σhf=	0.0547	m	SI CUMPLE
Σhf/q=	0.1161	m/l/s	
q=	-0.25429773	l/s	

ANILLO IV			Σhf<0.1
Σhf=	0.0434	m	SI CUMPLE
Σhf/q=	0.0564	m/l/s	
q=	-0.41476593	l/s	

ITERACIÓN V

ANILLO	RUTA		TRAMO	LONG(m)	Diam(pulg.)	Diam(m)	caudal q(lt/s)	coef. K	perdida de carga hf(m)	hf/Q	q(lt/s)	Caudal corregido Q(lt/s)
I	B	A	2	121.99	8	0.20	32.8402	284.96	0.5095	0.0155	-0.5400	32.3002
	A	F	6	135.86	6	0.15	2.8402	1288.24	0.0248	0.0087	-0.5400	2.3002
	B	G	7	135.86	6	0.15	-4.9608	1288.24	-0.0695	0.0140	-0.4466	-5.4074
	F	G	11	121.99	6	0.15	-12.1598	1156.73	-0.3285	0.0270	-0.5400	-12.6998
II	C	B	3	121.99	8	0.20	77.8010	284.96	2.5170	0.0324	-0.0934	77.7076
	B	G	7	135.86	6	0.15	4.9608	1288.24	0.0695	0.0140	0.4466	5.4074
	C	H	8	135.86	10	0.25	-84.7563	107.05	-1.1081	0.0131	-0.0145	-84.7708
	H	G	12	121.99	6	0.15	-27.1990	1156.73	-1.4589	0.0536	-0.0934	-27.2924
III	C	D	4	121.99	8	0.20	-77.4427	284.96	-2.4956	0.0322	-0.0789	-77.5216
	C	H	8	135.86	10	0.25	84.7563	107.05	1.1081	0.0131	0.0145	84.7708
	D	I	9	135.86	6	0.15	-5.7137	1288.24	-0.0903	0.0158	0.0044	-5.7093
	H	I	13	121.99	6	0.15	27.5573	1156.73	1.4947	0.0542	-0.0789	27.4784
IV	D	E	5	121.99	8	0.20	-31.7291	284.96	-0.4781	0.0151	-0.0833	-31.8123
	D	I	9	135.86	6	0.15	5.7137	1288.24	0.0903	0.0158	-0.0044	5.7093
	E	J	10	135.86	6	0.15	-1.7291	1288.24	0.0099	-0.0057	-0.0833	-1.8123
	J	I	14	121.99	6	0.15	13.2709	1156.73	0.3862	0.0291	-0.0833	13.1877

ANILLO I			Σhf<0.1
Σhf=	0.0653	m	SI CUMPLE
Σhf/q=	0.0653	m/l/s	
q=	-0.5399568	l/s	

ANILLO II			Σhf<0.1
Σhf=	0.0196	m	SI CUMPLE
Σhf/q=	0.1131	m/l/s	
q=	-0.09338536	l/s	

ANILLO III			Σhf<0.1
Σhf=	0.0169	m	SI CUMPLE
Σhf/q=	0.1153	m/l/s	
q=	-0.078899	l/s	

ANILLO IV			Σhf<0.1
Σhf=	0.0084	m	SI CUMPLE
Σhf/q=	0.0543	m/l/s	
q=	-0.08329105	l/s	

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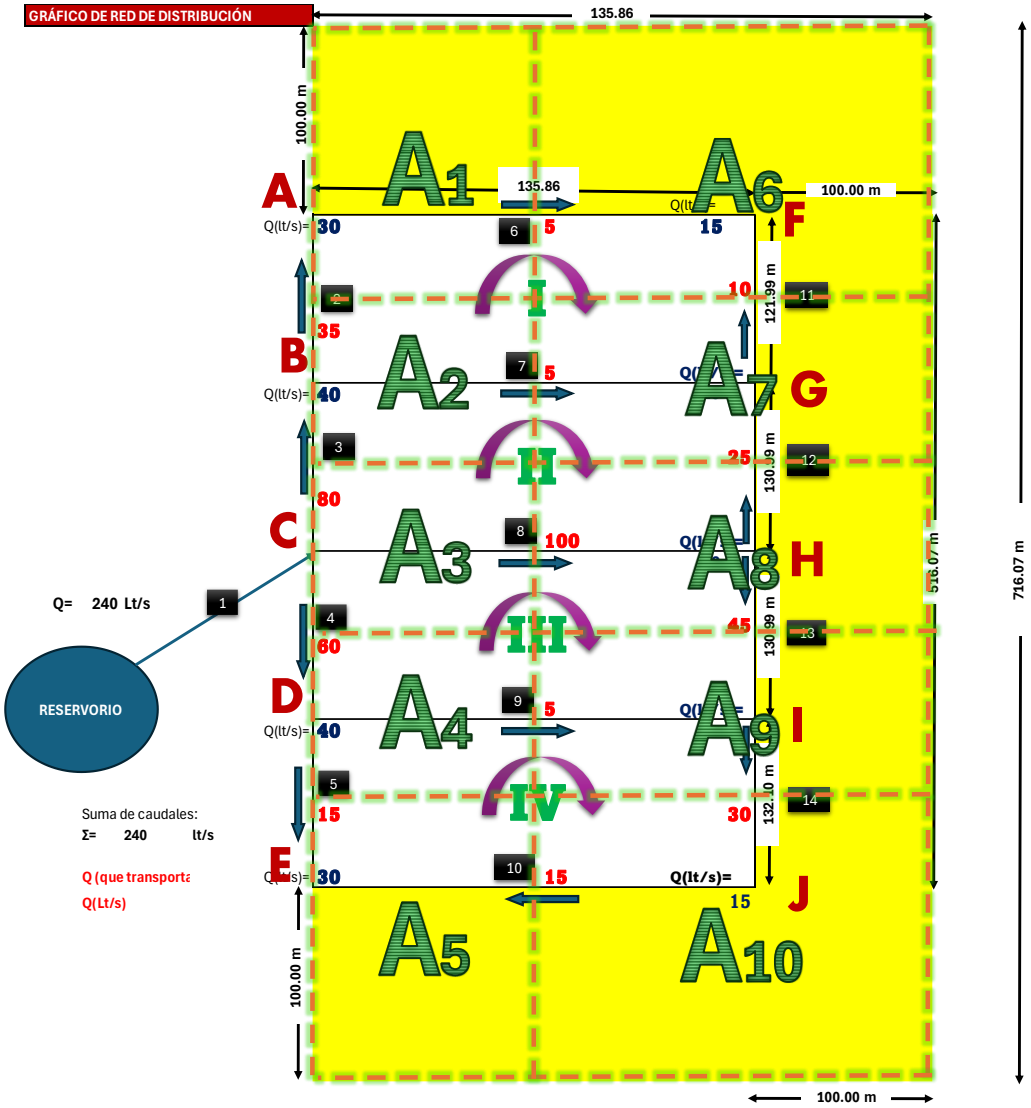
CÁLCULO CAUDAL UNITARIO PARA DISEÑO

GRÁFICA DE ÁREAS TRIBUTARIAS

Vivienda	6	hab/vivienda
i=	0.158	%
Po=	2760	hab
Periodo=	20	años

GRÁFICO DE RED DE DISTRIBUCIÓN



MANZANA	N° VIVIENDAS	ÁREA(m2)	Hab. Iniciales	Hab. Futuros	CAUDAL DE APORTE
M1	0	3744.8	0		0
M2	46	3744.8	276	286	24
M3	46	3744.8	276	286	24
M4	46	3744.8	276	286	24
M5	46	3744.8	276	286	24
M6	46	3744.8	276	286	24
M7	46	3744.8	276	286	24
M8	46	3744.8	276	286	24
M9	46	3744.8	276	286	24
M10	46	3744.8	276	286	24
M11	46	3744.8	276	286	24
M12	0	3744.8	0		0
TOTAL	**460**	**44937.6**	**2760**	**2860**	**240**

Qdis= 240 lt/s

CAUDAL UNITARIO Qu

Por vivienda	0.52173913	Lt/s/vivienda
Por área	0.005340739	Lt/s/área
Por área verde	0.17338424	Lt/s/área verde
Por area(total)	0.178724979	Lt/s/área
Por Habitante	0.083916084	Lt/s/habitante

Por tro lado: para AREA VERDE SEGÚN LA NORMA IS 010 la dotación de agua es:

2 Lt/dia/m2 ó 2.31×10^{-5} L/s/m^2

Entonces:

Mz.	ÁREA	DOTACIÓN
2	3744.8	0.00002315
PRODUCTO	0.17338424	Lt/s/área verde